

The cost of a mark: Newton’s vanishing sagitta and a floor of order $\hbar$

Abstract. Newton reads a force off a trajectory by letting the sagitta and the area between the inertial tangent and the curve go to zero, keeping their ratio to the time. We show that the same comparison, once it must be *recorded*, has a floor. For every protocol of marks, of any number and at any times, whose resolution and recoil obey $\delta_j \Delta_j \geq \kappa$, deciding between free and forced motion over a duration τ at error probability ϵ requires

$$\tau \Delta E = \frac{F^2 \tau^3}{2m} \geq 9 z_{1-\epsilon}^2 \kappa,$$

and for a mark realized by coupling the body’s position to a probe, the error operator and the impulse delivered are canonically conjugate, so $\kappa \geq \hbar/2$ exactly. The floor is about $12\hbar$ at five per cent error, and insertion of marks finer than $\tau_* = (9z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$ records free motion only. The historical claim is that Newton’s *Opticks* contains both factors of κ , one of them measured as the 1/89000th part of an inch, and that the single proposition joining them to a floor is an indeterminacy about the fits which Newton formulated the negation of. The distance between his system and a positive $\hbar$ is therefore one nameable proposition, and that proposition is Robertson’s inequality for his corpuscle.

Draft, 2026-09-17. Synthesis of the mark-floor, derivation, probabilistic and mark-cost notes, which hold the proofs in full. Two obligations remain before submission and are stated in §9. Exploratory; no ledger promotion.

1. The question

Galileo’s comparison is an inertial horizontal line against a falling parabola. With mass m , transverse force F , horizontal speed v and duration τ , the two paths are $q_I(t) = (vt, 0)$ and $q_F(t) = (vt, Ft^2/2m)$, and the region between them has

$$A = \frac{vF\tau^3}{6m}, \quad s(\tau) = \frac{F\tau^2}{2m}, \quad \Delta E = \frac{F^2\tau^2}{2m}, \quad \frac{3F}{v}A = \tau \Delta E. \quad (1)$$

The sagitta s , the area A and the product $\tau \Delta E$ all scale as powers of τ and vanish together. Newton’s Lemma X makes the first displacement quadratic in the time; Lemma XI, Corollaries 4 and 5, make the curved segment one third of the tangent triangle with both cubic in the time; Proposition VI reads the force off the limit $2ms/\tau^2$; and Proposition I builds an orbit as a polygon struck by impulses at its vertices, the number of vertices “augmented in infinitum”. The construction is designed so that the quantities in (1) disappear and their ratios survive.

Our question is what stops that refinement when the trajectory has to be recorded rather than contemplated, and whether the stopping point can be reached from materials Newton possessed.

The answer has four parts. The geometry alone supplies no floor (§2). Recording supplies one, which is set by a single number κ with the dimensions of action, and every protocol of marks faces the same bound in the same combination $F^2\tau^3/m$ (§3). Quantum kinematics fixes $\kappa \geq \hbar/2$ by an exact and elementary argument (§4). Newton’s optics supplies both factors of κ and denies the inequality between them (§§5–6). Newton also attempted a preface founded on the ancients, and the ancient arguments that bear on his own method are the ones he passed over; stated as premises about records they generate a hierarchy whose first rung is Zeno’s arrow and whose second is the comparison above (§7).

2. The geometry has no floor

Proposition 1. For every partition $0 = t_0 < \dots < t_n = \tau$ with steps τ_j and mesh $|\pi|$, the vertex impulse $F\tau_j$, the step sagitta $F\tau_j^2/2m$ and the step area $vF\tau_j^3/6m$ tend to zero with the

mesh; the areas sum to at most $vF\tau|\pi|^2/6m$; and $2ms(\tau_j)/\tau_j^2 = F$ exactly at every step.

The proof is immediate from (1). Berkeley's objection, that evanescent increments are "neither finite Quantities nor Quantities infinitely small, nor yet nothing", is answered inside the geometry by the limit ratio, which is what Newton's closing Scholium to Book I Section I provides. A floor must therefore come from the act of marking the curve, and the reading of "inserting a point" used here is physical: a point of the trajectory is inserted by making a mark on the body at a time.

3. Marks, and the floor they impose

Definitions. A **mark** at time t_j leaves a datum about the transverse coordinate. Its **resolution** δ_j is the spread of the resulting position estimate's error; its **recoil** Δ_j is the spread of the transverse impulse it delivers to the body. A protocol is a finite set of marks; it **decides** the comparison at error probability ϵ when some test on the full record does so with equal priors. A model satisfies the **mark trade-off** with constant κ when every available mark has $\delta\Delta \geq \kappa$.

The statistical model. Mark j returns $R_j = y(t_j) + \xi_j$ with ξ_j centred Gaussian of standard deviation δ_j , and delivers an impulse ι_j , centred Gaussian of standard deviation Δ_j , independently, so that

$$y(t) = y_0 + v_0 t + \theta P(t) + \frac{1}{m} \sum_j \iota_j (t - t_j)_+, \quad P(t) = \frac{Ft^2}{2m},$$

with $\theta = 0$ under the inertial hypothesis and $\theta = 1$ under the falling one, and y_0, v_0 unknown. Tests are invariant under the two unknowns, so the statistic is $u^\top R$ with $\sum_i u_i = 0$ and $\sum_i u_i t_i = 0$, and the optimal invariant test has error probability $\Phi(-d/2)$ with $d^2 = \sup(u^\top P)^2 / (u^\top \Sigma u)$ over such u .

Theorem 2. For every protocol satisfying the mark trade-off with constant κ ,

$$d^2 \leq \frac{2F^2\tau^3}{9m\kappa},$$

independently of the number of marks and of their times. Hence deciding at error probability ϵ requires

$$F^2\tau^3 \geq 18z_{1-\epsilon}^2 m\kappa, \quad \tau\Delta E \geq 9z_{1-\epsilon}^2 \kappa, \quad A \geq \frac{6z_{1-\epsilon}^2 v\kappa}{F},$$

with $z_{1-\epsilon} = \Phi^{-1}(1 - \epsilon)$.

Proof. The noise enters R in two ways: ξ_j affects R_j alone, and ι_j affects R_i for $i > j$ with coefficient $(t_i - t_j)/m$. Hence

$$u^\top \Sigma u = \sum_j \delta_j^2 u_j^2 + \sum_j \frac{\Delta_j^2}{m^2} S_j^2, \quad S_j = \sum_{i>j} u_i (t_i - t_j),$$

and the arithmetic-geometric mean inequality applied termwise gives $u^\top \Sigma u \geq \frac{2\kappa}{m} \sum_j |u_j| |S_j|$.

Introduce $N(w) = \sum_{i:t_i > w} u_i$ and $T(w) = \int_w^\tau N$. Since $\sum_i u_i = 0$, N vanishes for $w < 0$ and for $w \geq t_k$; T is continuous, piecewise linear, constant on $(-\infty, 0]$, zero on $[\tau, \infty)$, with $T' = -N$ jumping by u_j at t_j and $T(t_j) = S_j$. Two identities follow. Fubini and one integration by parts, using $P'(0) = 0$ and $T(\tau) = 0$, give

$$u^\top P = \int_0^\tau P' N = - \int_0^\tau P' T' = \int_0^\tau P'' T = \frac{F}{m} \int_0^\tau T;$$

and since T' has bounded variation with jumps u_j and no other variation, while TT' vanishes at both ends,

$$\sum_j u_j T(t_j) = \int_{\mathbb{R}} T dT' = - \int_{\mathbb{R}} T'^2 = -E, \quad E = \int_0^\tau T'^2.$$

Therefore $\sum_j |u_j| |S_j| \geq E$ and $u^\top \Sigma u \geq 2\kappa E/m$. Cauchy–Schwarz with $T(\tau) = 0$ gives $|T(w)| \leq \sqrt{\tau - w} \sqrt{E}$, hence $\int_0^\tau T \leq \frac{2}{3} \tau^{3/2} \sqrt{E}$, and

$$\frac{(u^\top P)^2}{u^\top \Sigma u} \leq \frac{(F/m)^{2/3} \tau^3 E}{2\kappa E/m} = \frac{2F^2 \tau^3}{9m\kappa}.$$

Finally $\Phi(-d/2) \leq \epsilon$ requires $d \geq 2z_{1-\epsilon}$. $\square$

Three features of the proof carry the paper’s claim to universality. The certificate u ranges over all invariant tests, so no monitoring scheme is privileged. The function T encodes the whole protocol in one object, and the bound depends on it only through the Dirichlet energy E , which cancels. And the exponent 3 on τ comes from a single Cauchy–Schwarz inequality against $\sqrt{\tau - w}$, which is where the combination in (1) originates.

Corollary 3 (insertion). Marks inside a window of duration τ' , used by themselves, record the force only if $F^2 \tau'^3 \geq 18z_{1-\epsilon}^2 m\kappa$. So insertion stops at

$$\tau_* = \left(\frac{18z_{1-\epsilon}^2 m\kappa}{F^2} \right)^{1/3},$$

and finer marks are consistent with free motion at the stated confidence. Newton’s polygon may be refined past τ/τ_* vertices, and its impulses remain in the geometry, though no record exhibits them.

A worst-case counterpart. If the marks instead confine position and impulse to intervals with certainty, the same three steps run with a separating hyperplane in place of the optimal test direction and give $F^2 \tau^3 > 9m\kappa$ for every protocol, with two marks sufficing above $36m\kappa$. That version has no quantum instance, since certain position confinement forces full-support momentum by Paley–Wiener; it is recorded here because it shows that the argument depends on the quadratic structure rather than on any probabilistic assumption.

4. The cost of a mark is the probe’s uncertainty product

Theorem 4. Let a mark be realized by the impulsive coupling of the body’s position to a probe, $U = \exp(-\frac{i}{\hbar} \lambda \hat{y} \hat{P}_A)$ with $[\hat{Q}_A, \hat{P}_A] = i\hbar$, the pointer $\hat{Q}_A$ being read afterwards. Then

$$U^\dagger \hat{Q}_A U = \hat{Q}_A + \lambda \hat{y}, \quad U^\dagger \hat{p} U = \hat{p} - \lambda \hat{P}_A,$$

so the error operator $\hat{N} = \hat{Q}_A/\lambda$ and the delivered impulse $\hat{D} = -\lambda \hat{P}_A$ satisfy $[\hat{N}, \hat{D}] = -i\hbar$ and

$$\kappa = \delta \Delta = \Delta \hat{N} \cdot \Delta \hat{D} = \Delta \hat{Q}_A \cdot \Delta \hat{P}_A \geq \frac{\hbar}{2}$$

for every probe state, pure or mixed.

Proof. With $A = \frac{i}{\hbar} \lambda \hat{y} \hat{P}_A$ the commutators $[A, \hat{Q}_A] = \lambda \hat{y}$ and $[A, \hat{p}] = -\lambda \hat{P}_A$ are central, so the Baker–Campbell–Hausdorff series terminates and gives the two displayed relations, while $\hat{y}$ and $\hat{P}_A$ are unchanged. Estimating the body’s position by $\hat{Q}_A/\lambda$ leaves the error $\hat{N} = \hat{Q}_A/\lambda$, and $[\hat{N}, \hat{D}] = -[\hat{Q}_A, \hat{P}_A] = -i\hbar$. Robertson’s inequality closes it. $\square$

The coupling strength cancels, so the cost is a property of the mark rather than of how hard it is made. The record and the recoil are themselves incompatible, since $[\hat{Q}_A + \lambda \hat{y}, -\lambda \hat{P}_A] = -i\hbar \lambda$, which is why no reading of the pointer determines the impulse. A deterministic probe, whose later position records fix its momentum, has $\kappa = 0$ and no floor; this is the exact point at which classical record models differ.

For Gaussian probe states and the linear dynamics of the comparison, the Wigner function is a probability density evolving classically, so the statistical model of §3 reproduces the quantum experiment exactly. With Theorem 4,

$$\tau \Delta E \geq \frac{9}{2} z_{1-\epsilon}^2 \hbar, \quad \tau_* = \left(\frac{9 z_{1-\epsilon}^2 m \hbar}{F^2} \right)^{1/3}, \quad (2)$$

about $12\hbar$ at $\epsilon = 0.05$ and $4.5\hbar$ at $\epsilon = 0.32$.

5. Both factors are in the *Opticks*

Newton’s optics supplies the two spreads whose product is κ , in the form of two least quantities of the probe.

A least length, measured. Book II Part III defines the interval of the fits as “the space it passes between every return and the next return” of the ray’s disposition to be reflected, and Proposition XVIII gives it: for the confine of yellow and orange passing perpendicularly into air, “the Intervals of their Fits of easy Reflexion are the 1/89000th part of an Inch”, that is about $0.285 \mu\text{m}$. A datum obtained through the fits repeats with that period, so it locates the corpuscle no better; Newton’s rings are a gauge of exactly this kind. This is $\Delta \hat{Q}_A$.

A least impulse, posited. Query 29 asks whether the rays of light are “very small Bodies emitted from shining Substances”, which gives each a transverse momentum scale. Newton has no way to measure it. This is $\Delta \hat{P}_A$.

Colour dependence, measured. Observation 13 of Book II Part I reports the ratio of the red to the violet interval as “greater than as 3 to 2, and less than as 13 to 8”, and “By the most of my Observations it was as 14 to 9”. Query 29 makes the violet corpuscles the least and the red the biggest. The product $\Delta \hat{Q}_A \Delta \hat{P}_A$ is therefore colour dependent on Newton-age evidence, and its universality across the spectrum is what Planck’s constant later supplies.

Inflexion. Query 1 asks whether bodies “act upon Light at a distance, and by their action bend its Rays”, most strongly at the least distance, which is the effect that limits a mark’s resolution when the beam is narrowed to improve it.

All five passages are held verbatim with line anchors in the source companion, from the fourth edition of 1730.

6. The junction, and the proposition Newton denied

Theorem 4 says what the missing premise is. In the mark-floor note it was stated as M3, that a mark fixing the position within δ leaves the delivered impulse undetermined within κ/δ . Theorem 4 identifies κ as the probe’s own uncertainty product, so

$$\text{M3 for Newton’s corpuscle} \iff \Delta \hat{Q}_A \cdot \Delta \hat{P}_A \geq \text{a positive constant},$$

which is Robertson’s inequality stated in Newton’s two quantities. With M3, §§3–4 give the floor and the insertion mesh from Newton’s own materials, with $\kappa = \Lambda p$ in place of $\hbar/2$; the identification $\Lambda = \lambda/2$, $p = h/\lambda$ makes $\Lambda p = h/2$, which differs from $\hbar/2$ by π .

Newton denied M3, and the denial is explicit rather than inferred. Proposition XII of Book II Part III states that every ray is “put into a certain transient Constitution or State, which in the progress of the Ray returns at equal Intervals, and disposes the Ray at every return to be easily transmitted”. The disposition is a determinate periodic property carried by the ray, so its fate at a surface is fixed by its phase, and what a prior record fails to supply is knowledge rather than determination. On that reading a probe’s position and momentum are both definite, later records can recover the impulse it delivered, $\kappa = 0$, and the comparison has no floor. The countermodel is explicit: let the corpuscle travel freely after the mark to a screen at distance

D , and two position records fix its transverse momentum to within $p(\delta + \delta_s)/D$, which vanishes as D grows.

The historical claim is therefore narrow and checkable. Newton’s optics contains both factors of the product that bounds the recorded comparison; his mathematics contains the comparison and takes it to zero; and the one proposition that would join them is an indeterminacy about the fits whose negation he asserted in print. The gap between the *Principia*’s vanishing sagitta and the *Opticks*’ finite interval of fits follows from that single commitment, and the separation of the two books is incidental to it. No counterfactual about what Newton might have discovered is needed, and none is offered.

7. Newton and the classics

What Newton attempted

In the early 1690s Newton drafted a set of Classical Scholia to Propositions IV to IX of Book III, arguing that the oldest philosophers had known universal gravitation and its inverse-square law, so that the *Principia* recovered a lost wisdom rather than announcing a novelty. He never printed them. He did pass material to David Gregory, whose *Astronomiae physicae et geometricae elementa* of 1702 opens with the programmatic sentence that lest the physics delivered here seem something new and unheard of in astronomy, *eandem vetustissimis Philosophis notam . . . ostendam*, I shall show it was known to the most ancient philosophers, naming Anaxagoras, Archelaus, Euripides and Pythagoras. That preface is the printed witness closest to the unpublished scholia, and the source companion holds its programmatic paragraph. The scholarship is Casini, “Newton: The Classical Scholia”, *History of Science* **22** (1984), 1–46, and Mosley, “The Origins and Sources of Newton’s Classical Scholia”, *Erudition and the Republic of Letters* **9** (2024), 171–217, both verified at metadata level here and both still to be read.

So Newton did attempt a preface founded on the classics, and the attempt was doxographic. He sought ancient authority for a law he had already proved, and the authority he sought was for the *conclusions* of Book III.

The ancient arguments he passed over

The ancient material that bears on the *method* of Book I is the debate over division and the cut, and it is absent from the scholia. Six positions, each held here in a primary witness:

- **Democritus’s cone** (Plutarch, *De communibus notitiis* 39). Cut a cone by a plane parallel to the base: are the surfaces of the two adjacent sections equal or unequal? If equal, the cone is a cylinder; if unequal, it is stepped. Chrysippus answers that the surfaces are neither equal nor unequal while the bodies are unequal, and Plutarch calls that a licence to write whatever comes to mind.
- **Zeno’s arrow** (Aristotle, *Physics* VI.9). At every now the flying arrow occupies a space equal to itself and so is at rest, and what is at rest at every now does not move. Aristotle’s diagnosis is that the argument assumes time to be composed of nows.
- **The Mohist endpoint** (*Mozi*, Canons and Explanations). Halving terminates at an endpoint that cannot itself be halved.
- **Liu Hui’s cutting** (commentary on the *Nine Chapters*, 263 CE). The finer the cutting the smaller the loss; cut and cut again until it cannot be cut, and the polygon coincides with the circle with nothing lost.
- **Hui Shi’s stick** (*Zhuangzi* 33). A stick one foot long, halved every day, is not exhausted in ten thousand generations.
- **The leap and the time-atom** (al-Shahrastani on al-Nazzam; Maimonides, *Guide* I.73). The interval is divisible, yet it is crossed by leaps; and for the kalam, time is composed of atoms.

- **The Vaiśeṣika arrow** (*Vaiśeṣikasūtra* 5.1.16–18, layered, c. 100 BCE–200 CE, attribution to Kaṇāda traditional). The arrow’s particular conjunctions are *ayugapat*, non-simultaneous, and that is the ground of the plurality of its motions; impulsion causes only the **first** motion, and each later one is caused by the *saṃskāra* deposited by the motion before it, until the impression is spent and gravity takes over.
- **Sound produced from sound** (*Vaiśeṣikasūtra* 2.2.36–37 in the Candrānanda recension, 2.2.31 in the vulgate). Sound arises from conjunction, from disjunction, and from sound, and is non-eternal, so what reaches the ear is a later member of a generated chain. Vātsyāyana’s *Nyāyabhāṣya*, within the window at about 400 to 450 CE, argues the point from observation: the axe-blow is still heard at a distance after the axe-wood contact has ceased.
- **No motion at all** (Vasubandhu, *Abhidharmakośabhāṣya* ad IV.2, c. 350–450 CE). *na gatir yasmāt saṃskṛtaṃ kṣaṇikam*, there is no motion, because the conditioned is momentary; everything conditioned “is destroyed in the very place where it arose”, so its passage to another place is impossible. Zeno concludes this and treats it as absurd; Vasubandhu concludes it and accepts it.
- **A medium that permits motion** (Umāsvāti, *Tattvārthasūtra* 5.17–18, c. 2nd–5th c. CE). *gatisthityupagrahau dharmādharmayor upakārah*: the assisting of motion and of rest is the function of *dharma* and *adharma*. These substances push nothing; they are the standing condition without which motion could not occur, and space is given the separate office of accommodation.
- **Sound as spreading waves** (Diogenes Laertius VII.158, reporting Stoic doctrine). Hearing occurs when the air between is struck, “a vibration which spreads spherically and then forms waves and strikes upon the ears, just as the water in a reservoir forms wavy circles when a stone is thrown into it”. The same water-ring image carries the Vaiśeṣika *vīcī-santāna*, so the model is attested independently in both traditions.
- **Continuity that fails below sense** (Epicurus, *Letter to Herodotus* 61–62, in Diogenes Laertius X). Atoms travel at equal speed through the void, and in aggregates they “move in different directions in times so short as to be appreciable only by the reason, but frequently collide until the continuity of their motion is appreciated by sense”; the assumption that continuity persists below observation “is not true in the case before us”.
- **Least partless bodies** (Diodorus Cronus, reported at Sextus Empiricus, *Pyrrhoneion Hypotyposesis* III.32). Listed in the doxography of material principles between the atoms of Democritus and Epicurus and the unjointed masses of Heraclides. Diodorus argued from those minima that “a thing never is moving, but it has moved”, the Greek twin of Vasubandhu’s *na gatiḥ* three centuries earlier; the formula is at *Adversus Mathematicos* X and is cited here without being read.
- **The voice as spherical waves** (Vitruvius, *De architectura* V.3.6–7, c. 25 BCE). The voice is propelled “by an infinite number of circles similar to those generated in standing water when a stone is cast therein”, but “whereas the circles in water only spread horizontally, the voice, on the contrary, extends vertically as well as horizontally”, so the water-ring image is corrected into a spherical wave by argument from the disanalogy.
- **The sling, applied to an orbit** (Plutarch, *De facie in orbe lunae* 923C–D, c. 100 CE). “The moon is saved from falling by its very motion and the rapidity of its revolution, just as missiles placed in slings are kept from falling by being whirled around in a circle”, with 923D adding that “each thing is governed by its natural motion unless it be diverted by something else”.

Lemmas X and XI perform exactly the operation these positions dispute. The scholia cite

the ancients for what Book III concludes and leave them silent on how Book I proceeds.

The last four entries differ in kind from the rest, and they matter most here. The Greek and Chinese items are paradoxes about division, which set a problem. The Indian entries answer it, and they answer it three different ways: the arrow's flight is many motions carried by an impression (Vaiśeṣika), or it is no motion at all because each thing perishes where it arose (Vasubandhu), or it is possible only because a medium stands ready to permit it (Umāsvāti). Each also carries a stopping rule for division: the Nyāya *paramāṇu* than which nothing is smaller, the directional-parts reductio by which a touched atom would have parts, and the Jain atom that has no space-points at all. The Vaiśeṣika sūtras in particular state a **positive theory of propagation**: motion is a succession of numerically distinct events, each carried to the next by a quantity the previous one deposits, and transmission through a medium is a chain in which what arrives is a later member than what was sent. That is the structure Theorem 2 quantifies. Its protocol-universality holds because the signal is a phase-space path whose total variation is unchanged by subdivision, which is the modern form of the claim that impulsion supplies only the first motion while the impression carries the rest. An ancient author had therefore already framed propagation as a bounded succession rather than a continuous traversal, and Newton's classical preface reached for none of it.

The hierarchy the ancient premises generate

Read as premises about records rather than about geometry, the cone and the arrow are the first members of a sequence whose next member is Newton's comparison. The apparatus of §3 covers all of them; what changes is the order of the signal against the nuisance it must beat.

Theorem 5. In the model of §3, with every mark obeying $\delta_j \Delta_j \geq \kappa$:

- (i) *The arrow.* Distinguishing rest from uniform motion at speed v , with the initial position unknown, requires

$$m v^2 \tau \geq 8 z_{1-\epsilon}^2 \kappa, \quad \text{that is} \quad \tau E_{\text{kin}} \geq 4 z_{1-\epsilon}^2 \kappa.$$

- (ii) *Newton.* Distinguishing uniform motion from constant force, with the initial position and velocity unknown, requires $F^2 \tau^3 / m \geq 18 z_{1-\epsilon}^2 \kappa$, which is Theorem 2.
- (iii) *The cone.* Measuring a static taper requires nothing. With no recoil to propagate, the deflection grows without bound as marks accumulate, so a static solid has no floor.

Proof. For (i) the nuisance is the constant, so u ranges over vectors with $\sum_i u_i = 0$ alone. With $P(t) = vt$ the identity of §3 reads $u^\top P = v \sum_i u_i t_i = v T(0)$, and $T(\tau) = 0$ with Cauchy–Schwarz gives $|T(0)| = |\int_0^\tau T'| \leq \sqrt{\tau} \sqrt{E}$. The noise bound $u^\top \Sigma u \geq 2\kappa E/m$ needs only $\sum_i u_i = 0$ and so is unchanged. Hence $d^2 \leq v^2 \tau E m / (2\kappa E) = m v^2 \tau / (2\kappa)$, and $d \geq 2 z_{1-\epsilon}$ gives the claim. For (ii) the nuisance is the linear space, $u^\top P = \frac{F}{m} \int_0^\tau T$, and the proof of Theorem 2 applies. For (iii) the marks leave the object unchanged, so $\Sigma = \text{diag}(\delta_j^2)$; with N marks of resolution δ at each of two heights separated by Δh , the taper ϑ gives $d^2 = N \vartheta^2 \Delta h^2 / (2\delta^2)$, unbounded in N . $\square$

The two dynamical cases share one invariant. If the signal is the n -th order departure P with $P(0) = \dots = P^{(n-1)}(0) = 0$, the quantity the theorem bounds below is

$$m (P^{(n)})^2 \tau^{2n-1} \gtrsim \kappa,$$

which has the dimensions of action for every n : at $n = 1$ it is $m v^2 \tau$, twice the kinetic energy times the duration, and at $n = 2$ it is $F^2 \tau^3 / m$, twice $\tau \Delta E$. Newton's comparison is the second rung of a ladder whose first rung is Zeno's.

Why the arrow and not the sling

One feature of the list needs saying. Almost every ancient entry is rectilinear or static, while the case Newton uses to **define** centripetal force is the sling: Definition V has the stone whirled about in a sling endeavouring to recede from the hand, and names the force that “retains it in its orbit” after the planets “perpetually drawn aside from the rectilinear motions, which otherwise they would pursue”.

The exception is Plutarch, and it is a pointed one. *De facie* 923C has the moon kept from falling “just as missiles placed in slings are kept from falling by being whirled around in a circle”, which is Newton’s apparatus applied to Newton’s case a millennium and a half earlier. The inference runs the other way: for Plutarch the whirling *prevents* the fall, so speed sustains; for Newton the cord *pulls the stone inward*, and what it diverts is a straight line that would need no cause. The same object supports opposite accounts, and what separates them is which motion is taken to be free. Plutarch’s next clause, that “each thing is governed by its natural motion unless it be diverted by something else”, is as near as the passage comes, with natural motion still doing the work inertia later does. Newton had read him: the Classical Scholia name Plutarch among the ancients said to have known the doctrine of gravitation, so the sling was available in a text he was mining for authority while writing the propositions it models.

Theorem 5 explains why the exception stayed isolated, and Aristotle supplies the other half of the reason. *De caelo* I.2 makes uniform circular motion the natural motion of the aether, needing no cause, which is the exact inverse of Definition V. For the heavens, then, curved motion was the default and straight motion the anomaly, so the sling illustrates a conclusion already held rather than posing a problem. The claim that the ancients treat rest as natural holds below the moon and is incomplete above it; the reference is at metadata level here. The ancient question is the first rung, whether the thing moves at all, with position as the only nuisance. The sling is the second rung, whether a force acts, with uniform motion as a further nuisance, and that rung cannot be stated until straight-line motion is held to need no account. Ancient physics largely does not hold that, so the sling has no work to do in it. The Vaiśeṣika comes closest by charging only the first motion to impulsion, and stops short of making a straight continuation free and a curved one costly.

Indian sources do supply one whirled object, and use it for a third purpose again. The firebrand circle, *alātacakra*, appears in Vasubandhu (Pradhan 189.23–24) to argue that because contact with the parts is successive, the awareness of a whole is really of the parts, and at Pradhan 33.9 with *āśuvṛtṭyā*, by rapid action. That is an argument about sampling, and it is the ancient form of what Theorem 5 says about records: below the mesh a succession is indistinguishable from the continuous thing it mimics. The companion note sets this out.

What each ancient premise buys

- **Zeno.** His conclusion holds of records: below $\tau_{\text{arrow}} = 8z^2\kappa/(mv^2)$ every mark is consistent with rest. Aristotle’s diagnosis explains why, since a now carries no record and every datum is a window.
- **The Vaiśeṣika arrow and the chain of sounds.** These supply the positive half the Greek material lacks. Theorem 5’s ladder is their statement made quantitative: the succession is real, each step carries a bounded quantity to the next, and reading any one step costs κ . The sound chain is the case where an ancient author asserts a finite propagation time and argues it from an experiment, which is the premise Newton’s determinate fits deny for light.
- **Vasubandhu.** His conclusion is what the theorem says about *records* rather than about the world. Below the mesh every mark is consistent with rest, so no recorded instant exhibits motion and travel is never displayed, only inferred from enough instants together. The theorem declines his further step from the records to the world, and Theorem 5(i)

measures exactly how many instants are needed.

- **Epicurus.** His is the closest ancient statement to what the theorems say about records. Motion presents itself as continuous to sense and is a succession below it, and he names the fallacy of extrapolating the observed continuity downward. He draws the boundary where sense fails; Theorem 5 draws it at a mesh computed from the mark cost, which is the difference between a threshold that is reported and one that is derived.
- **Umāsvāti.** The Jain medium is the one ancient category with no counterpart in the theorem, and its absence is informative. Theorems 2 and 5 need no medium: what they price is the mark, not the motion. A doctrine on which motion requires a permitting substance predicts nothing about the cost of observing it, which is the respect in which this entry sets a limit on how far the reconstruction reaches.
- **Democritus and Chrysippus.** The dilemma about adjacent sections dissolves with resolution alone, and case (iii) shows it needs no action floor. Adjacent recorded sections are equal, and the inequality of the bodies is recovered cumulatively, which is Chrysippus's answer given a definite sense. This corrects a heuristic recorded earlier in this programme, that the cone is the arrow rotated into Euclidean time: the static problem lacks back-action, and back-action is the entire source of the floor.
- **The Mohists and Liu Hui.** They are right about records. Cutting stops, and the endpoint is the mesh $\tau_* = (18z^2 m \kappa / F^2)^{1/3}$, which is Corollary 3.
- **Hui Shi.** He is right about geometry, which is Proposition 1: the halving never ends, and nothing in the figure resists it.
- **Al-Nazzam.** The recorded trajectory below the mesh is his leap. The interval stays divisible, and the crossing is exhibited only in finitely many steps.
- **The kalam time-atom.** Denied. The mesh depends on F , on m and on the confidence demanded, so it is a dynamical resolution rather than a universal atom of time. A floor on action coexists with a continuum of instants, and Theorem 5 shows the mesh moving as the force changes.

The section's claim about Newton is therefore double. He looked to the ancients for authority and found the wrong ancients, taking the doxography of gravitation while passing over the dispute about division that his own Lemmas turn on. And the premise that would have converted that dispute into a theorem, an indeterminacy in the least parts of light, is one the atomists supply in the form of a least part and one he denied in the single place where he had a measurement.

8. What the theorem does not give

Preparations of unbounded extent. Theorem 2 concerns protocols that mark the trajectory. A protocol that prepares once, waits and measures once faces a different bound. In that case the two hypotheses differ by the phase-space displacement $(-Ft^2/2m, Ft)$, whose components have product exactly $\tau\Delta E$, and Mandelstam–Tamm in the geometric form of Anandan and Aharonov (PRL **65**, 1697, 1990), with Helstrom's discrimination bound (*Quantum Detection and Estimation Theory*, 1976), gives

$$\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \geq 1 - 2\epsilon, \quad \tau\Delta E \geq \frac{(1 - 2\epsilon)^2 \hbar^2}{4A},$$

for an apparatus whose position and momentum spreads are bounded by L and P with $A = LP \geq \hbar/2$; and this holds for every finite adaptive protocol of instruments, because the displacement is a phase-space path whose total variation does not grow when it is subdivided. The bound is tight: two narrow packets separated by $\pi\hbar/(F\tau)$ decide the comparison with error probability of order $\tau\Delta E/\hbar$, for arbitrarily small $\tau\Delta E$, at the price of an apparatus whose extent diverges. So the floor of order $\hbar$ requires either that the trajectory be marked or that the laboratory be bounded, and a non-Gaussian preparation of unbounded extent evades it.

Prior art. The combination $F^2\tau^3 \gtrsim m\hbar$ is the standard quantum limit for detecting a force on a free mass (Braginsky and Khalili, *Quantum Measurement*, 1992; the canonical statement is Caves, Thorne, Drever, Sandberg and Zimmermann, *Rev. Mod. Phys.* **52**, 341, 1980), and its status has been disputed since Yuen’s objection: Caves defended it (PRL **54**, 2465, 1985) and Ozawa exhibited a measurement breaking it for free-mass position (PRL **60**, 385, 1988). That non-Gaussian preparations beat the limit, which is §7’s counterexample, is stated as a principle by Giovannetti, Lloyd and Maccone (*Science* **306**, 1330, 2004). The contribution here is universality over protocols with an explicit constant, the exact mark cost of Theorem 4, and the identification of the premise. The inequality itself is not new.

The classical theorem belongs to a literature it does not cite. Worst-case recovery of a linear functional from noisy linear data is optimal recovery in the sense of Micchelli and Rivlin (1977), and the apparatus series this programme built, together with Theorem 1’s worst-case form, are results of that kind; information-based complexity (Traub, Wasilkowski and Woźniakowski, 1988) is the same setting. The invariant-test step of Theorem 2 is standard (Lehmann and Romano). A referee from either community will ask why these go unmentioned, and the honest answer is that the theorems were derived without them.

Instruments outside the momentum-transfer class. Theorem 4 holds where $[\hat{N}, \hat{D}] = -i\hbar$. Its coupling is von Neumann’s (1932), its inequality Robertson’s (1929), and its ancestor the Heisenberg microscope (1927); Bohr and Rosenfeld (1933) and Araki and Yanase (1960) are where limits of this kind from field measurability and from conservation laws are set. Ozawa’s counterexamples to the naive Heisenberg product concern instruments where that commutator differs, and a general class would need the proved calibration relation of Busch, Lahti and Werner (PRL **111**, 160405, 2013) or an argument of another kind. No error–disturbance relation is used above.

9. Obligations before submission

Two, both on the history side, and both stated so that a reader can see what the present draft rests on.

1. **Historiography, to be read rather than cited.** On the *Opticks* side, Shapiro’s *Fits, Passions, and Paroxysms* (1993) on the theory of fits and how determinate Newton meant it, which is where §6’s central claim must be tested, and Sabra’s *Theories of Light from Descartes to Newton* (1981) for the context of Query 29. On the *Principia* side, Guicciardini’s *Reading the Principia* (1999) and his *Isaac Newton on Mathematical Certainty and Method* (2009) for what the limit arguments of §2 were for, De Gandt’s *Force and Geometry* (1995) directly on the sagitta, and Bertoloni Meli’s *Thinking with Objects* (2006) on the sling and the pendulum as objects to think with. On the scholia, McGuire and Rattansi, “Newton and the ‘Pipes of Pan’” (1966), beside Casini and Mosley. On the ancient comparison, Sorabji’s *Time, Creation and the Continuum* (1983), which treats Zeno, Diodorus Cronus, Epicurean minima and kalām atoms together; von Rospatt (1995) on momentariness; Dhanani (1994) on kalām atomism and the Indian influence question; and Lloyd and Sivin’s *The Way and the Word* (2002) as the methodological standard that protects §7 from the charge of naive parallelism. Clagett (1959) for impetus, where Philoponus’ *rhōpē* is the Western counterpart of *samskāra*.
2. **Editions.** The *Opticks* passages come from a transcription of the 1730 fourth edition and should be cited from that printing; the *Principia* passages should be cited from Cohen and Whitman in place of the Motte text used here.

Three mathematical items remain open and are not obligations of the paper: extending Theorem 4 beyond the momentum-transfer class; the general force law, for which the proof of

Theorem 2 already pairs $\int_0^\tau P''T$ against the Dirichlet energy and so bounds a norm of P'' ; and closing the constants between the worst-case form, the statistical form of Theorem 2 and the single-shot form of §7.

10. Consequence for STATE

This is the synthesis the goal asked for: the Planck gap established with Newton-age arguments and their modern equivalents, in one document whose historical and technical halves are load-bearing for each other. The foundations content is Theorems 2 and 4 with §7's honest positioning; the history content is §§5–7, resting on the source companions; the junction is the identification of M3 with Robertson's inequality for the corpuscle. STATE's queue reduces to §9's obligations and the three open mathematical items.